



Nova Class

Smart Attendance Management for College Classrooms

Nova Class is a web-based attendance platform designed to replace manual attendance registers with a controlled digital workflow. It supports teachers, students, and department HOD/admin users, with separate access and role-specific dashboards.

App Type	College attendance portal
Core Users	Students, Teachers, Department HOD/Admins
Main Goal	Mark, monitor, and export attendance accurately
Live App	https://smart-attendance-ecru-nu.vercel.app/login
Local URL	http://127.0.0.1:3001/

What Nova Class Does

Nova Class manages attendance through a live classroom session. A teacher creates a session for a department, section, subject, period, and date. The system generates a short live token, students submit that token, and the application records the attendance result against the correct class session.

- Tracks attendance by class session, subject, period, and date.
- Prevents duplicate attendance by allowing one mark per student per session.
- Supports automatic absent marking when a teacher closes a session.
- Shows students their subject-wise attendance percentage and attendance history.
- Gives HOD/admin users department-specific student lists, monthly attendance percentage, and CSV export.

User Roles

Role	What They Can Do
Student	Log in, submit the live attendance token, use biometric verification where enabled, view attendance history
Teacher	Create live attendance sessions, refresh tokens, view session attendance, manually override attendance
HOD/Admin	Access only their own department, filter students by year and month, view monthly attendance percentage

Department Admin Model

Each HOD/admin account represents one department, such as IT, CSE, Civil, or Mechanical. The admin dashboard is scoped to that admin department, so an IT HOD sees IT students and reports, while a CSE HOD sees CSE data. This separation helps keep department records clean and avoids cross-department access in normal use.

Monthly Attendance Percentage

The admin report shows attendance as a percentage for the selected month. The calculation is based on the number of present records divided by total class sessions for that student in the selected department/year/month. The CSV export includes serial number, roll number, name, attendance percentage, present count, and total session count.

Main Workflow

1. Teacher logs in and creates a class attendance session.
2. Nova Class generates a short live token for that session.
3. Student logs in and submits the token during class.
4. The system verifies role, enrollment, token validity, and duplicate submission rules.
5. Attendance is stored in the database as present or absent.
6. Teacher closes the session; missing enrolled students can be marked absent.
7. HOD/admin reviews department attendance and downloads monthly reports.

Technology

Nova Class is built with Next.js and TypeScript for the web application, Supabase for authentication and database storage, PostgreSQL row-level security concepts for data protection, Zod for request validation, and Vercel for deployment.

Security And Accuracy

- Role-based routing separates student, teacher, and admin dashboards.
- Admin APIs check that the logged-in user has the admin role before returning department data.
- Attendance submit logic checks enrollment and prevents duplicate attendance for the same session.
- Rate limiting helps reduce abuse of the attendance submission endpoint.
- Biometric/passkey support is available for student verification when configured on HTTPS domains.

In Simple Words

Nova Class is like a digital attendance register for a college. Teachers open attendance for a class, students mark themselves present using a secure live token, and HODs can later check department-wise attendance by year and month. The system reduces manual work and gives clear attendance percentages and downloadable reports.

Generated for the Nova Class attendance app.